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Analytica Chimica Acta 579 (2006) 209–216
ANALYTICA CHIMICA ACTA
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New reliable Raman collection system using the wide area illumination (WAI) scheme combined with the synchronous intensity correction standard for the analysis of pharmaceutical tablets

Minjung Kim [a], Hoeil Chung [a,*], Youngah Woo [b], Mark Kemper [c]

[a] *Department of Chemistry, Hanyang University, Seoul 133-791, Republic of Korea*
[b] *Korea Institute of Toxicology, Daejon 305-343, Republic of Korea*
[c] *Kaiser Optical Systems, 371 Parkland Plaza, Ann Arbor, MI 48103, United States*

Received 8 June 2006; received in revised form 14 July 2006; accepted 14 July 2006
Available online 25 July 2006

Abstract

A novel and reliable Raman collection system for the non-destructive analysis of pharmaceutical tablets has been proposed. The main idea was to develop and utilize the wide area illumination (WAI) scheme for Raman collection to cover a large surface area (coverage area: 28.3 mm^2) of solid tablet to dramatically improve the reliability in sample representation and the reproducibility of sampling due to less sensitivity of sample placement with regard to the focal plane. Simultaneously, the effective and synchronous standard configuration using isobutyric anhydride was harmonized with the WAI scheme to correct the problematic variation of Raman intensity from the change of laser power. To verify the quantitative performance of the proposed scheme, the compositional analysis of active ingredient in naproxen tablet has been performed. The average sample composition was successfully represented by using the WAI scheme compared to the conventional scheme with a much smaller illumination area. After the intensity correction using the non-overlapping peak of isobutyric anhydride standard and area normalization, a partial least squares (PLS) model was developed using an optimized spectral range and the concentrations of naproxen in tablets were accurately predicted. The prediction accuracy was not sensitive to changes in laser power or tablet position within ±2 mm. Additionally, the prediction accuracy was repeatable without systematic drift or need for re-calibration.
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Keywords: Raman spectroscopy; Tablet analysis; Naproxen; Non-destructive analysis

1. Introduction

The compositional analysis of pharmaceutical tablets has been extensively studied using Raman spectroscopy since it can provide selective and diverse information on molecular structure and chemical composition [1–3]. Since Raman intensity is quite dependent on laser power, sample presentation and optical collection configuration, it has been only adoptable with a great care for the analysis of tablets of simple composition where a peak ratio method can be adopted as a calibration method.

To boost Raman spectroscopy as a tool that can be widely utilized for the analysis of tablets in the pharmaceutical industry, two major issues should be critically addressed. The first is the inaccurate compositional representation of sample by the Raman spectrum due to the fact that the laser illuminates only on tiny spot (50–500 μm) to collect spectral information. The typical approach to represent average sample composition is rotation of the sample during Raman collection to cover a wide surface area [4–7]. A recent publication showed the coverage of wider area (19.6 mm^2) by sample rotation led to improved calibration performance [5]. However, to achieve fast and simple tablet analysis for applications such as those for PAT (Process Analytical Technology), the rotation of the sample is practically impossible and an alternative Raman collection scheme must be employed.

The other requirement for quantitative Raman measurements is the need for internal or external standards to correct Raman intensity fluctuations related to the variation in incident laser power. Internal standard methods cannot be utilized for the analysis of tablets since addition of an internal standard is not

* Corresponding author. Tel.: +82 2 2220 0937; fax: +82 2 2299 0762.
E-mail address: hoeil@hanyang.ac.kr (H. Chung).

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doi:10.1016/j.aca.2006.07.036

practical. Therefore, it is necessary to use a Raman collection scheme with an external standard method that should collect the Raman spectrum of both a sample and standard in a synchronous fashion [8–10]. If the collection of an external standard is not simultaneous, it would significantly increase the total analysis time.

To improve the problems of inaccurate sample representation and intensity variation in Raman tablet analysis, we have proposed a novel Raman collection system using a wide area illumination (WAI) scheme that applies a laser beam to a sample in a circular fashion with a 6 mm diameter (area: 28.3 mm^2). In this manner, the Raman spectrum can represent the average sample composition correctly and measurements can be accomplished quickly without rotation of the sample. Simultaneously with the WAI scheme, we have installed a layer of isobutyric anhydride (10 mm thickness) as an external standard in front of the tablet to correct for Raman laser intensity. The resulting Raman spectrum was the summation of spectral features from both the sample and standard, allowing the non-overlapping isobutyric anhydride peak to be used to correct intensity variation.

To evaluate the validity of this Raman collection scheme, a quantitative model for the compositional analysis of naproxen tablets was developed. Naproxen is a widely-used anti-pyrrhetic agent. Raman spectra were directly collected for naproxen tablets and a partial least squares (PLS) [11–13] regression was performed to quantify the naproxen content. By acquiring Raman spectra with proper representation of sample composition and effectively correcting Raman intensity using the external standard, the concentrations of naproxen in tablets were accurately determined.

2. Experimental

2.1. Sample preparation

The total weight of the naproxen tablets in this study was 200 mg. The tablets were white and composed mainly of 68.5% (137 mg) of naproxen as the active ingredient, and 19.4% and 10.0% of lactose and maize starch as excipients. The minor excipients included were polyvinyl pyrrolidone as a binder, cross-linked sodium carmellose as a disintergrant, and magnesium stearate as a lubricant. The naproxen concentration for each tablet was expressed as the percentage relative to the amount of naproxen in a normal tablet (137 mg). A tablet containing 137 mg was designated as 100% naproxen. Seven different concentrations (85, 90, 95, 100, 105, 110 and 115%) of naproxen tablets were prepared. Three tablets were prepared at each concentration and a total of 21 tablets were used for this study. Fig. 1 shows the dimensions of the naproxen tablets prepared and the molecular structure of naproxen.

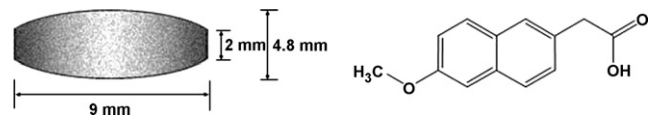

Fig. 1. Drawing of naproxen tablet and molecular structure of naproxen.

2.2. Raman spectral collection and data processing

The wide area illumination (WAI) scheme used in this study has been developed by Kaiser Optical Inc. (Ann Arbor, MI, USA). The schematic diagram is shown in Fig. 2. The WAI scheme provides unique advantages over the conventional Raman collection scheme to improve the performance of quantitative Raman analysis. It can improve the reliability of Raman measurements by significantly increasing surface coverage area. The reproducibility of sampling is also improved due to less sensitivity of placement of the sample with regard to the focal plane. Finally, there is less possibility for sample damage with use of the WAI scheme due to the reduced energy density of the laser spot incident to the sample.

The excitation laser (785 nm, diode laser) was introduced and magnified to form a circular illumination area of 6 mm diameter (area: 28.3 mm^2) to cover a large sample area. Conventional dispersive and FT-Raman have an illumination spot size from 50 to 500 μm; therefore, the surface coverage area of the WAI scheme is 40–900 times larger. The scattered radiation was then collected by an array of 50 optical fibers and delivered to the CCD detector. The radiation from the optical fibers was dispersed via a holographic transmission grating and a combined signal was integrated over the total area illuminated. With this scheme, the Raman spectrum could be acquired over approximately 44% of naproxen tablet surface. If the conventional scheme was used, only 0.05–1.2% of the surface area can be covered, which could possibly lead to poor quantitative results due to the failure of the Raman measurement to represent the whole concentration of tablet correctly. This issue has been a major hindrance for

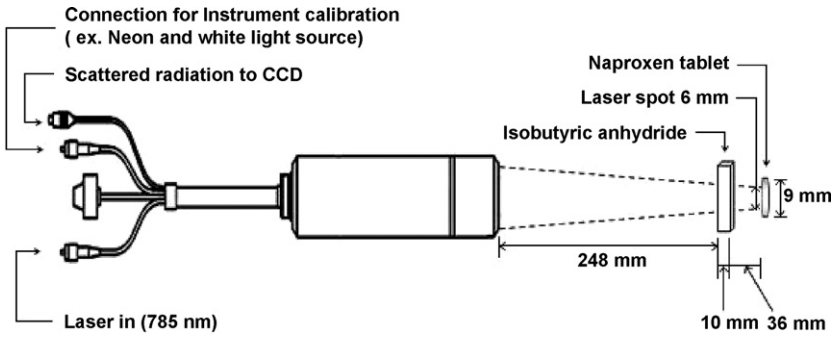

Fig. 2. Schematic diagram of the wide area illumination (WAI) scheme.

Raman to be adopted for the analysis of pharmaceutical tablets as well as many solid samples.

An extra optical fiber is present for the connection of neon and white light sources when wavelength and intensity calibration are necessary. To correct Raman intensity variation from an uncontrollable change of laser power, isobutyric anhydride sealed in a 10-mm thick rectangular quartz cell was installed at 36 mm in front of the tablet. Therefore, the resulting Raman spectrum was the summation of spectral features from the naproxen tablet and isobutyric anhydride. A non-overlapping band of isobutyric anhydride could be utilized to correct laser power variation to improve the quantitative calibration model.

Triplicate spectra were collected for each tablet. Forty-two spectra from 14 tablets were used as the calibration set to build PLS calibration models and 21 spectra from seven independent tablets were used as the validation set to verify these models. Both the calibration and validation sets included full range variation of naproxen concentration (85–115%). Raman spectra were collected using an exposure time of 3 s and 32 scans (96 s total time).

All the calculations including baseline correction, intensity normalization and partial least squares (PLS) regression, were accomplished using Matlab version 7.0 (The Math-Works Inc., MA, USA).

3. Results and discussion

3.1. Raman spectral features

Since the naproxen tablets were composed of several different components, it was initially required to examine the spectral features of each component. For more accurate Raman measurements, it is advantageous for the spectral features of naproxen (an active ingredient) to be distinct compared to those of other components. Fig. 3 presents the Raman spectra (1900–200 cm^{-1}) of each component in the naproxen tablet. The powder form of each component was used for data collection with the WAI without isobutyric anhydride standard. Among these materials measured, naproxen shows the most distinct spectral features and exhibits the greatest spectral density; i.e., there are several characteristic peaks throughout the whole spectral range, while other ingredients exhibit relatively less characteristic peak. Because the level of naproxen is greater than the other components in the tablets used for this study, the resulting Raman spectra should contain relatively strong peaks from naproxen with its characteristic spectral features such as the C=C double bond at 1630 cm^{-1}. However, because the spectra representing the tablets will show the overlapping spectral features from each component weighted by their levels, multivariate calibration methods such as the partial least squares (PLS) regression method are required to resolve the overlapping spectral features and finally quantify the naproxen concentrations.

From a practical viewpoint, the major drawback for the use of Raman spectroscopy for the analysis of pharmaceutical tablets is the inaccurate compositional representation of the sample from the illumination of the laser on a small spot as discussed. To improve sample representation, the physical rotation of a sample during data collection has been used to cover a wide surface area [4–7]. However, this approach is practically difficult to be employed in routine situations where high throughput sample analysis is required.

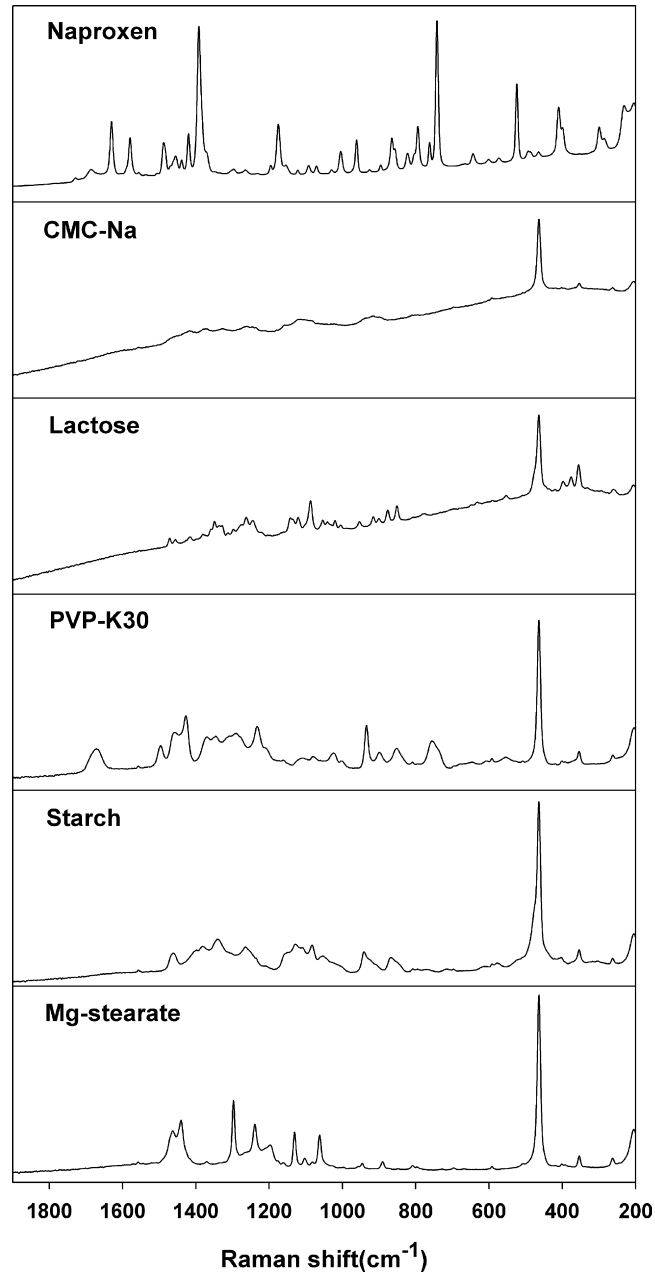

Fig. 3. Raman spectra (1900–200 cm^{-1}) of each component in naproxen tablet.

To investigate whether sample representation can be improved over the conventional scheme, Raman spectra of a 100% naproxen tablet were collected by the WAI scheme and a conventional small-spot scheme (non-contact Raman probe with a 10-mm focal length) to compare the resulting spectral features. Spectra were collected by randomly illuminating the laser on five different spots on the tablet surface using both approaches. The resulting spectra from two different spectral ranges (1040–880 and 480–320 cm^{-1}) are shown in Fig. 4. The Raman spectra collected by the WAI scheme show the same spectral feature in both

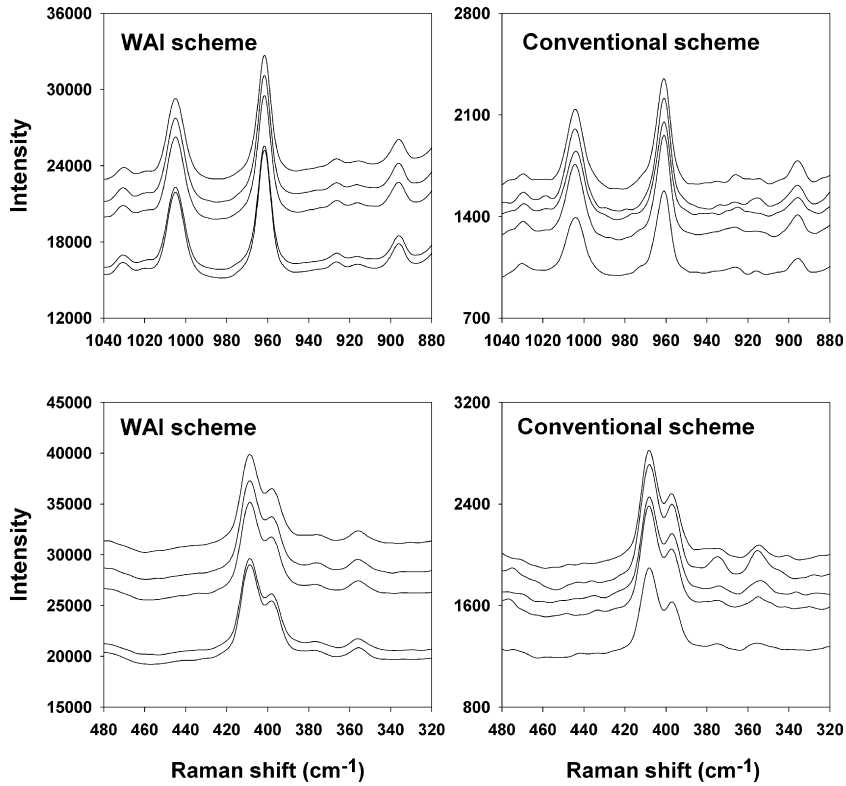

Fig. 4. Raman spectra collected by randomly illuminating laser on five different spots on 100% naproxen tablet using WAI scheme and conventional scheme (Raman probe with 10 mm focal length) to compare the resulting spectral features. Two different spectral ranges (1040–880 and 480–320 cm^{-1}) are shown.

spectral ranges even though the baselines vary vertically due to the slight difference in surface roughness. By illuminating the laser onto a large surface (area: 28.3 mm^2), the average composition of the tablet is effectively represented by the acquired Raman spectrum without sample rotation. When the conventional scheme is used, the resulting Raman spectra are fairly different from each other, even though the spectra are acquired from the same tablet. Since the size of the laser illumination spot in the conventional scheme is much smaller (approximately 100 μm in diameter) relative to the naproxen tablet (9 mm in diameter), a resulting Raman spectrum is only representative of a localized area and does not satisfactorily represent the total composition of the naproxen tablet. The same result was obtained for the analysis of ambroxol tablets as described in the previous publication [4]. Fig. 4 clearly demonstrates the importance of wide area coverage for the Raman analysis of pharmaceutical tablets. Improved reliability of the WAI scheme suggests quantitative analysis of tablets can be accomplished due to the superior compositional representation of the tablet.

Raman spectroscopy is a single beam technique and there is, therefore, no normalization for source fluctuations as there are in other techniques such as NIR, UV and FT-IR. It is commonly known that the variation of the laser power fluctuation should be compensated for reliable quantitative Raman measurement. Therefore, a suitable standard method accompanying the WAI scheme could also be benefit for quantitative analysis. In this study, we uniquely employed a synchronous external standard configuration by positioning the standard material in front of the naproxen tablet as shown in Fig. 2. Using this configuration, the Raman spectrum of a sample and external standard can be collected in a synchronous fashion and the resulting Raman spectrum is the summation of the spectral features from the tablet and external standard. To utilize an external standard to correct Raman intensity, it is required to have at least one band of the standard that does not overlap the sample bands. The intensity of such an isolated band from a standard can compensate for laser power variation. Isobutyric anhydride was chosen for this purpose in this study because the band at 1810 cm^{-1} (carbonyl stretching) was not overlapped with peaks from the naproxen tablet as shown in Fig. 5. For comparison, the Raman spectrum of naproxen and the molecular structure of isobutyric anhydride are shown in the same figure. In addition, isobutyric anhydride is chemically and physically stable (b.p.: 182 °C) for use over a long period of time. The non-overlapping 1810 cm^{-1} band was used to follow and correct laser power variation. This configuration can also be used for the effective external standard method for the analysis of other tablets, as long as a proper standard material possessing a non-overlapping peak with the bands from a given tablet can be found.

3.2. Intensity correction and normalization

Fig. 6(a) shows the untreated Raman spectra collected from tablets containing seven different concentrations of naproxen (two tablets per concentration and three spectra per tablet were collected to produce six spectra per concentration) with the presence of the isobutyric anhydride standard. The isolated standard band of isobutyric anhydride at 1810 cm^{-1} is clearly observed

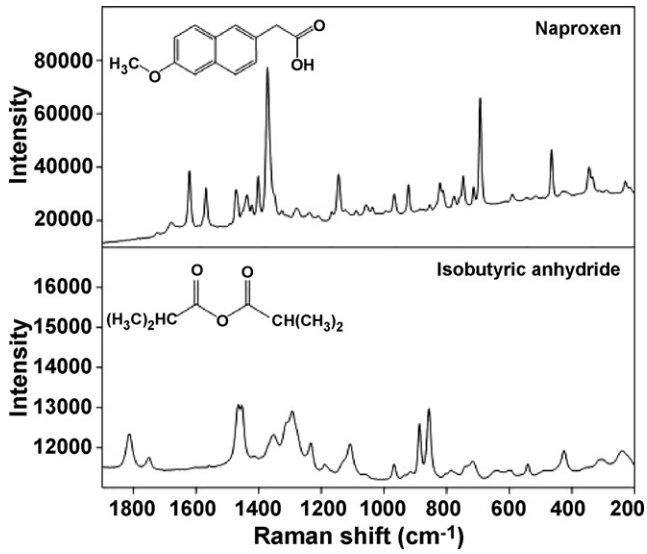

Fig. 5. Raman spectra of naproxen and isobutyric anhydride. The non-overlapping band of isobutyric anhydride 1810 cm^{-1} can be utilized to correct intensity variation from laser power change.

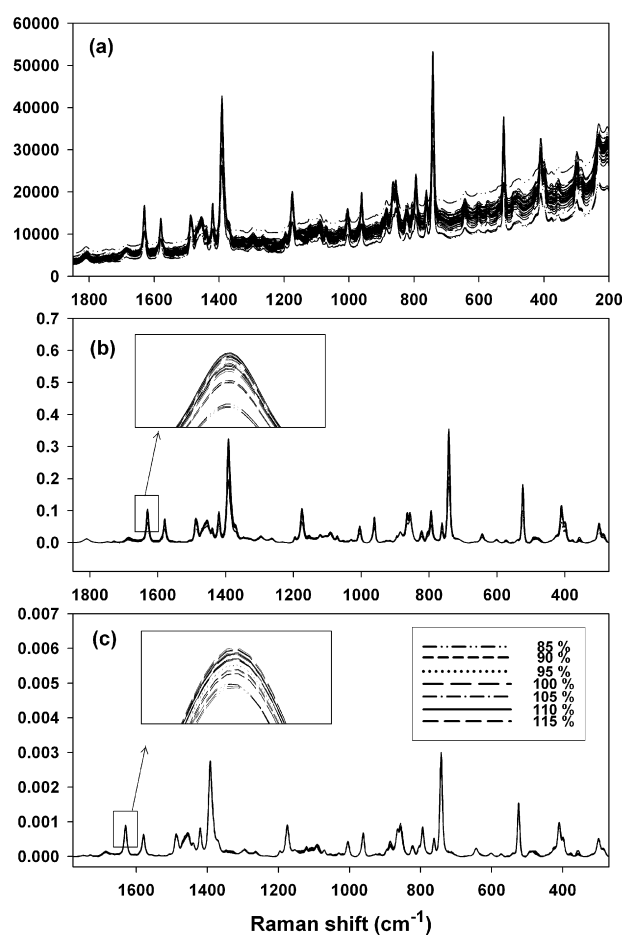

Fig. 6. Untreated Raman spectra (42 spectra) collected from seven different concentrations of naproxen tablets (a), the corresponding intensity corrected (b) and normalized spectra (c).

and a slight non-linear baseline is exhibited superimposed on the spectral features of the naproxen tablet. Initially, the underlying baseline was corrected using a 7-point (1847, 1777, 1218, 986, 689, 343 and 270 cm^{-1}) non-linear correction method. Next, the band area of isobutyric anhydride in the 1847–1777 cm^{-1} range was calculated and each sample spectrum was divided by the calculated area to compensate for any Raman intensity variations from occasional laser power fluctuation. The results are shown in Fig. 6(b) and the 1630 cm^{-1} band is enlarged in the same figure. The data indicate that the band intensity increases with the naproxen concentration and the Raman intensity is effectively corrected using the standard; however, the Raman intensity changes somewhat for tablets with the same concentration.

Since the surface contour of the naproxen tablet is curved, the laser illumination area can be varied when the tablet position is slightly changed. The change of illumination area can introduce a slight variation in the resulting Raman intensity. This should be removed or minimized for reliable quantitative performance. For this purpose, the area under the 1770–270 cm^{-1} range was calculated and each sample spectrum was divided by the calculated area to minimize the Raman intensity variation from the slight difference in the tablet position during the measurement. The resulting spectra are shown in Fig. 6(c). The 1630 cm^{-1} band is also enlarged. It is clear that the 1630 cm^{-1} band varies with the increase of naproxen concentration. Much more reproducible spectra are generated for the same naproxen tablet. To develop partial least squares (PLS) calibration models, the intensity-corrected and normalized spectra shown in Fig. 6(c) were used.

3.3. PLS calibration and range optimization

The use of the whole spectral range in PLS models could be advantageous since all the information can be utilized. However, this might not always be effective when the spectral features of a given analyte are broadly overlapping with those from other components. The possibility that an optimal spectral range that can yield improved calibration results should be investigated. Several chemometric strategies such as Moving Window PLS (MW-PLS) [14–16] and interval PLS (iPLS) [17] to find the best calibration range have been reported in attempt to find the best calibration range. We have used MW-PLS to find the optimal spectral range. The concept is to perform PLS at all possible combinations of spectral ranges to find the best spectral range that yields a minimum error. For this purpose, the window size (number of spectral points) was varied from 50 to 1700 cm^{-1} with increments of 10 cm^{-1}. With a given window size, the window was moved from low to high wavenumber with 50 cm^{-1} steps to cover the whole spectral range (1770–270 cm^{-1}). To perform MW-PLS, the original calibration set of 42 spectra was divided into 28 spectra to construct a new calibration set and 14 spectra to format a new validation set. With each spectral range, PLS models were developed using a new calibration set and the resulting standard error of prediction (SEP) from the new validation set was evaluated. These new calibration and validation sets were used only for the MW-PLS optimization procedure. A

Table 1
Overall PLS calibration results for determination of naproxen concentration

	Spectral range (cm^{-1})	SECV (%)
Full range	1770–270	0.96 (2)
Optimized range	740–290	0.53 (2)

Numbers in parentheses correspond to the number of factors used. The results from the use of optimized and full spectral ranges are compared.

total of 6615 spectral ranges with different window sizes were investigated, and the optimal ranges providing the best SEP were identified.

A PLS procedure was performed again in the identified optimal spectral range using the original calibration set (42 spectra from 21 samples) and the corresponding SECV (Standard Error of Cross Validation) was calculated. For the comparison, the whole spectral range (1770–270 cm^{-1}) was also used for PLS. The results are summarized in Table 1. The numbers in parentheses correspond to the number of factors used. As shown, the SECV is considerably improved using the optimized spectral range, 740–290 cm^{-1}. It is difficult to provide a clear rationale for the optimized spectral range; however, the naproxen spectral features in the optimized range are less overlapped with those from other components compared to the 1770–740 cm^{-1}, as shown in Fig. 2. Therefore, it is expected that the lesser degree of spectral overlap has lead to the improved calibration performance. The corresponding correlation plot resulting from the PLS calibration using the optimized range (740–290 cm^{-1}) is shown in Fig. 7. As shown, the naproxen concentrations of the tablets were accurately calibrated with an SECV of 0.53%.

3.4. Prediction performances in diverse situations

To further demonstrate the prediction robustness of the proposed Raman system, we have generated Raman spectra under several situations that may be encountered in routine analysis. First, the gradual fluctuation of laser power occurs commonly.

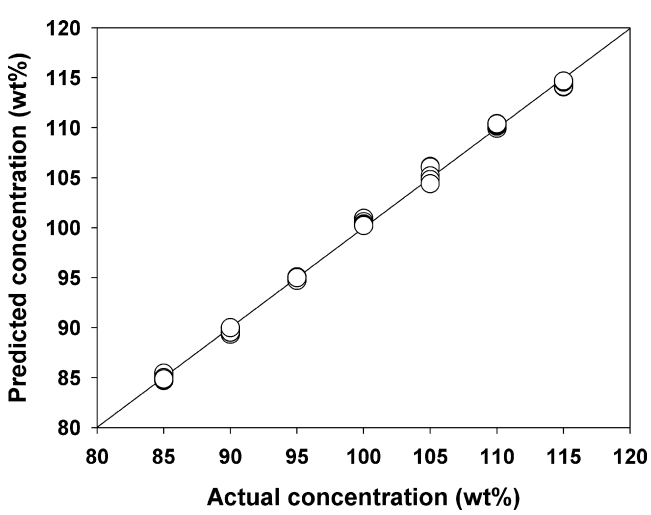

Fig. 7. Concentration correlation plot resulted from PLS calibration using the optimized range (740–290 cm^{-1}).

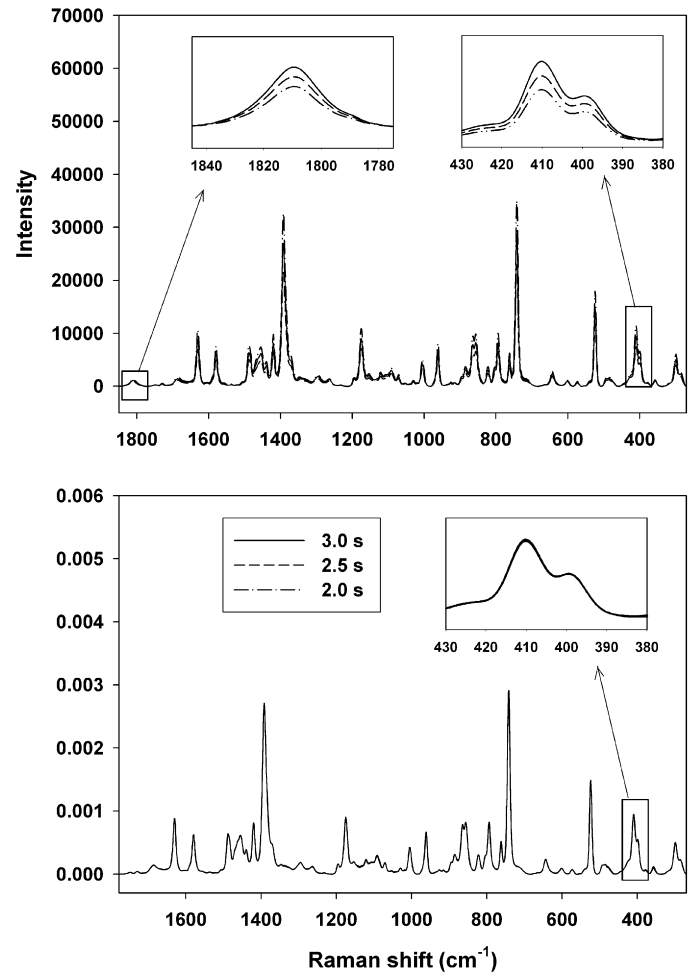

Fig. 8. Baseline corrected Raman spectra of 100% naproxen tablet collected with three different exposure times (a) and all 21 spectra (intensity corrected and normalized) collected at three different exposure times (b).

Prediction accuracy should be maintained under such circumstances. To simulate this situation, we intentionally decreased the exposure time to 2.5 and 2 s, and collected Raman spectra using the seven tablets in the prediction set. In addition, Raman spectra were also collected with an exposure time of 3 s (the initial instrumental setting). At each exposure time, seven spectra from seven different tablets (one spectrum per tablet) were collected. In total, 21 spectra with three different exposure times were acquired and predicted using the developed PLS model as described in Table 1. Fig. 8(a) shows baseline corrected Raman spectra of a 100% naproxen tablet collected with three different exposure times. As clearly shown in the enlarged plot, the intensities of the naproxen band at 410 cm^{-1} and the isobutyric anhydride band at the 1847–1777 cm^{-1} range both increase with exposure time. Fig. 8(b) shows the corresponding intensity corrected and normalized spectra. The resulting Raman intensities of the tablet with the 100% naproxen concentration are similar to each other. The intensity variation from different exposure times was effectively corrected using the non-overlapping isobutyric anhydride band for normalization. The resulting SEP was 0.53% and the corresponding concentration correlation plot is shown in Fig. 9(a). The prediction accuracy was successfully maintained

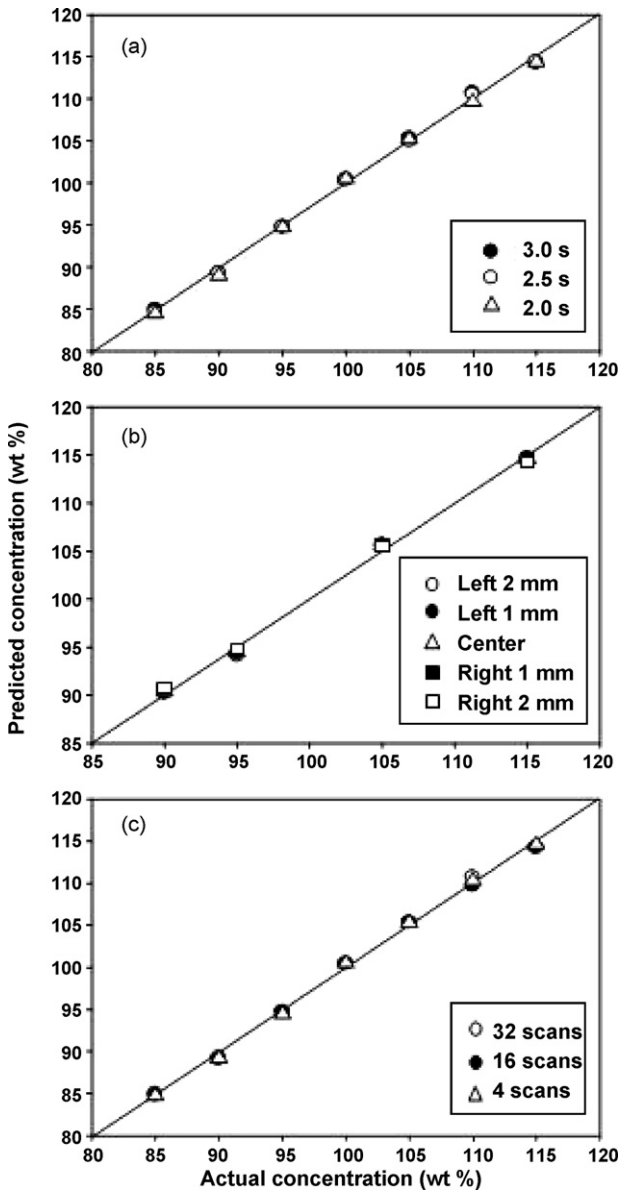

Fig. 9. Concentration correlation plots for prediction under the variation of laser exposure time (a), tablet position (b) and scan number (c).

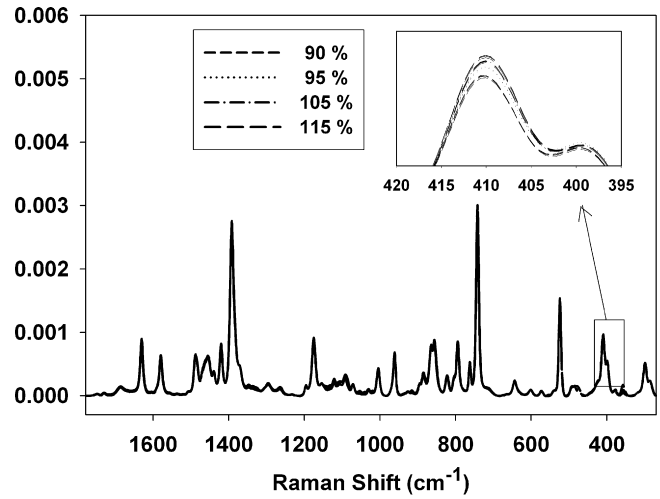

Fig. 10. Raman spectra (intensity corrected and normalized) collected at five different positions in each tablet. Four different concentrations of naproxen tablets were used (total 20 spectra).

and the naproxen concentrations were accurately predicted even under the variation in the Raman intensity.

An additional important issue is the change of tablet position with each Raman measurement since the exact control of tablet position might not be realistic in routine or in-line analysis. If the Raman measurement is sensitive and strongly influenced by the sample position, it is practically difficult to be adopted for routine use. To address this issue, Raman spectra of four different tablets were collected deliberately changing the position relative to the laser between each measurement. Initially, the spectrum was collected at the center (original position) as before. Then, the tablets were moved to the left by 1 and 2 mm from the center and the corresponding Raman spectra were collected at each position. The same procedure was repeated by positioning a tablet to the right by 1 and 2 mm. This variation in sample position is relatively large when the size of the tablet is considered (9 mm in diameter). Overall, five spectra were collected at five different positions for each tablet and a total of 20 spectra (from four tablets) were predicted by the developed PLS calibration model using the optimized spectral range. The resulting raw spectra at five different positions were very similar each other. This is a unique advantage of the WAI scheme combined with external standard method for the analysis of tablets with curved surfaces. Since the large surface area was illuminated, the variation of illuminated area did not vary significantly when tablet position was changed and the resulting Raman intensity was not strongly influenced. Fig. 10 shows all 20 spectra (intensity corrected and normalized) collected at the five different tablet positions. The variation of the band at 410 cm^{-1} is enlarged in the figure. The increase in Raman intensity is clearly observed with the increase of naproxen concentration and the intensities of the same naproxen tablet are similar to each other even when the tablet position was changed as described. On the contrary, if the conventional scheme with a small illumination spot was utilized for the same purpose, the variation of tablet position would introduce a much larger variation in Raman intensity. Therefore, the prediction performance was stable with a variation in the tablet position of ±2 mm. The resulting SEP was 0.54%. The corresponding concentration correlation plot is shown in Fig. 9(b).

To analyze one tablet, 96 s were required (3 s exposure and 32 scans). To make the proposed Raman scheme applicable for PAT (Process Analytical Technology) applications where rapid analysis is essential, the analysis time should be lessened. To investigate the feasibility of faster analysis, we used seven tablets representing the full concentration range. We collected Raman spectra of each tablet by changing the scan numbers to 32, 16 (acquisition time: 48 s) and 4 (acquisition time: 12 s). A total of 21 spectra with different scan numbers were acquired. The naproxen concentrations were predicted by the optimized PLS model described in Table 1. The resulting SEP is 0.52% and the corresponding concentration plot is shown in Fig. 9(c). The signal-to-noise is adequate to compensate for such fluctuations because of the summed signal collected by the WAI configura-

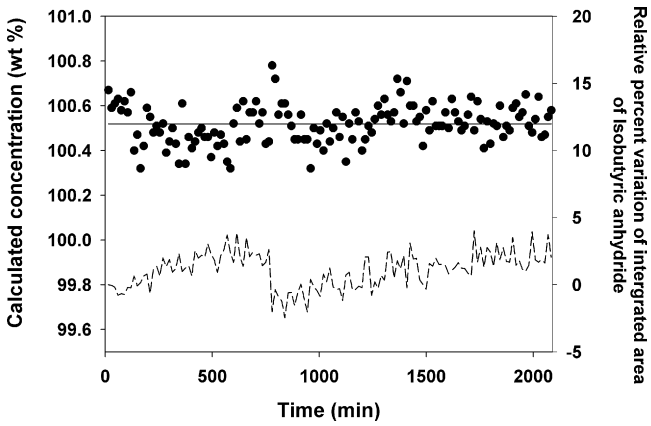

Fig. 11. Prediction results for 100% naproxen tablet collected over 24 h at 15 min interval and relative percent variation of integrated area of isobutyric anhydride in the 1847–1777 cm^{-1}.

tion. Therefore, the minute increase of noise by decreasing the scan number did not dramatically influence the overall signal-to-noise ratio of the spectrum and the prediction performance was not affected. Based on this result, it is possible to decrease the analysis time below 10 s as long as a PLS model is developed accordingly.

Finally, the long-term repeatability of the WAI scheme was evaluated by continuously collecting the spectra of one 100% naproxen tablet over 38 h at 15 min intervals, thus examining laser power variation over 1 day. Fig. 11 shows the prediction results for all of the 139 spectra. In addition, the integrated area of isobutyric anhydride in the 1847–1777 cm^{-1} was calculated for the first sample and relative percent variations of integrated areas for subsequent spectra were obtained. The result is displayed in the same figure. The prediction results were repeatable with a standard deviation of 0.1% and no systematic trend was observed.

4. Conclusion

The WAI scheme combined with a synchronous external standard was successful to determine the concentration of active ingredient in naproxen tablets. The WAI scheme was unaffected with respect to reproducibility by the surface curvature. From a practical viewpoint, the Raman scheme presented in this study can be used in the field of pharmaceutical analysis and could improve the efficiency of measurements in quality assurance situations. Additionally, the proposed scheme can be utilized for the identification of pharmaceutical raw materials where samples are in diverse forms. For such cases, a fast, non-destructive analytical method is preferable. We are also developing a modified WAI Raman scheme that can directly measure samples in diverse glass or plastic bottles, such as pharmaceutical syrups and sterile liquids by utilizing the long focal length (≈25 cm) of the system.

Acknowledgements

This work was supported by the Korea Science and Engineering Foundation (Grant No. R-14-2002-004-01000-0). The authors thank Kaiser Optical Systems for the collaboration and the supply of instrumentation.

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